data. To cover harm risks in real-world user requests, we include prompts from LMSYS-CHAT-1M [42] and WILDCHAT [41] datasets, both of which capture in-the-wild human-LLM interactions, with harm labels assigned based on the OpenAI Moderation API. Additionally we add subsamples of existing annotator-written safety datasets to further enhance coverage. This includes prompts from HH-RLHF [5] and the Anthropic Red-Teaming dataset [11], including the subsets that make up AegisSafetyTrain [13] and SAFETY-TUNED LLAMAS [7].

3.1.2 Compliance and Refusal Construction

LLM response generations. For our synthetic adversarial and vanilla prompts, we generate matched refusal and compliance responses. We submit each prompt to a suite of LLMs,[3] along with a prompt suffix instructing the LLM to refuse or to comply with the prompt. This provides us with a set of refusal and compliance candidates.

GPT-4 complex response generations. We additionally use GPT-4 to generate items capturing a set of targeted response types in observed challenge areas for refusal classification. We conduct error analysis on a classifier trained on an early prototype of WGTRAIN, and identify several categories of nuanced responses that the classifier mislabels. The majority of these categories are compliances containing caveats, warnings, etc. We then construct one-shot prompts for GPT-4 to generate synthetic responses with similar properties to enhance coverage of complex responses.[4]

3.1.3 Filtering, Auditing, and Sampling

We further filter responses created through open LMs by assigning labels for each of our three target tasks using GPT-4 and recategorizing items that fail to match intended labels. We then audit the quality of these GPT-4 labels by sampling 500 items and collecting human annotations as in §3.2. This audit confirms the high quality of the GPT-4 labels, which agree with voted annotator labels on 92%, 82%, and 95% of items for prompt harm, response harm, and refusal labels, respectively. After our filtering and audit, we sample from these sets the following numbers of prompts along with matched refusal and compliance responses: 6,062 vanilla harmful prompts, 2,931 vanilla benign prompts, 4,489 adversarial harmful prompts, and 4,339 adversarial benign prompts, for 35,642 prompt+response items. We also include prompt-only items of 10,451 vanilla harmful prompts, 3,086 vanilla benign prompts, 11,289 adversarial harmful prompts, and 11,411 adversarial benign prompts.

We sample from our other data sources (on which filtering/audit is not needed) as follows: from complex response, 1,167 each of refusal, compliance, and prompt-only items; from IN-THE-WILD data, 944 harmful prompts and 944 benign prompts; and from annotator-written data, 7,361 benign and 2,130 harmful prompts.[5] IN-THE-WILD and annotator-written items contain only prompts.

3.2 WILDGUARDTEST: A High-Quality Human-Annotated Test Moderation Dataset

We build WILDGUARDTEST (WGTEST) to address the limitations of existing moderation tool evaluations discussed in §2, including harm classification on adversarial inputs and refusal detection.

To construct WGTEST, we start with a test split of 1,725 prompt-response pairs from the synthetic vanilla and adversarial data as described in §3.1. We then collect annotations from three independent annotators for each prompt-response pair on prompt harmfulness, response refusal, and response harmfulness. Fleiss Kappa [10] scores are 0.55, 0.72, and 0.50 for the three tasks, indicating moderate to substantial agreement. We apply majority voting to obtain gold labels, removing items that fail to reach at least two-way agreement.[6] Finally, we run a prompted GPT-4 classifier on the dataset and manually inspect items on which the output mismatches the chosen annotator label, to further audit the ground-truth labels. See Table 10 in Appendix A.5 for the WGTEST label breakdowns and harm category splits according to our risk taxonomy, and Appendix E for more annotation details.

[3] OLMo-7B-Instruct, GPT-3.5, Vicuna-7b-v1.5, Llama3-8B-Instruct, Mistral-7B-Instruct-v0.2, and Dolphin variants of `dolphin-2.9.1-llama-3-8b`, `dolphin-2.8-gemma-7b`, and `dolphin-2.8-mistral-7b-v02`.

[4] Most items take user prompts by XSTest category from the vanilla benign prompt set, while for a few categories the user prompts were also synthetically generated. See Appendix A.5 for all categories and prompts.

[5] See Appendix A.7 for further breakdowns.

[6] Lack of agreement is possible because we include an "unsure" answer choice.

Table 3: F1 scores (%) on prompt and response harmfulness in existing public benchmarks.

Model	Prompt Harmfulness (F1)						Response Harmfulness (F1)				
	ToxiC	OAI	Aegis	SimpST	HarmB	**Avg.**	HarmB	S-RLHF	BeaverT	XST	**Avg.**
Llama-Guard	61.6	75.8	74.1	93.0	67.2	74.4	52.0	48.4	67.1	82.0	62.4
Llama-Guard2	47.1	76.1	71.8	95.8	94.0	77.0	77.8	51.6	71.8	90.8	73.0
Aegis-Guard-D	70.0	67.5	84.8	100	77.7	80.0	62.2	59.3	74.7	52.8	62.3
Aegis-Guard-P	73.0	74.7	82.9	99.0	70.5	80.0	60.8	55.9	73.8	60.4	62.7
HarmB-Llama	-	-	-	-	-	-	84.3	60.0	77.1	64.5	71.5
HarmB-Mistral	-	-	-	-	-	-	87.0	52.4	75.2	72.0	71.7
MD-Judge	-	-	-	-	-	-	81.6	64.7	86.7	90.4	80.9
LibrAI-LongFormer	-	-	-	-	-	-	62.1	50.6	67.1	70.9	62.7
BeaverDam	-	-	-	-	-	-	58.4	72.1	89.9	83.6	76.0
OAI Mod. API	25.4	79.0	31.9	63.0	9.6	41.8	20.6	10.1	15.7	46.6	23.2
GPT-4	68.3	70.5	84.4	100	100	84.6	86.1	67.9	86.1	91.3	82.0
WILDGUARD	70.8	72.1	89.4	99.5	98.9	**86.1**	86.3	64.2	84.4	94.7	**82.4**

XSTEST-RESP: XSTest with responses. We also create XSTEST-RESP by extending XSTest with model responses to directly evaluate moderator accuracy for scoring models on a real safety benchmark. We use the same LLM suite employed above for response generation to create completions for the prompts in XSTest [30], randomly sample from these outputs, and then collect human annotations similarly to WGTEST for response harm and refusal labels. After filtering for inter-annotator agreement, we have a total of 446 prompts for response harmfulness (368 harmful responses, 78 benign responses) and 449 prompts for refusal detection (178 refusals, 271 compliances).

3.3 Training WILDGUARD with WILDGUARDTRAIN

With WGTRAIN, we instruction-tune WILDGUARD using `Mistral-7b-v0.3` [19] as a base model.[7] We design a unified input and output format to capture the three tasks, providing the user prompt and model response as inputs, and training the model to output elements corresponding to all three tasks. See more training details in Appendix B.

4 Evaluating WILDGUARD Against Existing LLM Safety Moderation Tools

We show that WILDGUARD substantially improves performance relative to existing LLM safety moderation tools on all of the *prompt harm*, *response harm*, and *refusal detection* tasks, across existing benchmarks and our newly introduced WGTEST evaluation sets.

4.1 Evaluation Setups

Evaluation benchmarks. We test WILDGUARD and relevant baselines on ten publicly-available safety benchmarks, as well as our WGTEST across all three tasks. For public benchmarks on prompt harmfulness, we use ToxicChat [23], OpenAI Moderation [24], AegisSafetyTest [13], SimpleSafetyTests [36], and HarmBenchPrompt [25]. For public benchmarks on response harmfulness, we use HarmBenchResponse [25], SafeRLHF [9], BeaverTails [18], and XSTEST-RESP[8]. The only evaluation for refusal detection other than WGTEST is XSTEST-RESP, which we discuss in §2. Table 14 in Appendix C shows statistics of each benchmark. We report F1 scores for all evaluations.

Existing Safety Moderation Models. As in Section 2, we test four LM-based moderation tools trained for prompt and response harmfulness detection: Llama-Guard, Llama-Guard2, Aegis-Guard-Defensive and Aegis-Guard-Permissive [16, 26, 13]. We test five additional models for response harmfulness classification: BeaverDam [18], LibrAI-LongFormer-harm [38], MD-Judge-v0.1 [22], and two Harmbench classifiers from Mazeika et al. [25] (HarmBench-Llama, HarmBench-Mistral). For refusal detection, we test LibrAI-LongFormer-ref [38]. See Appendix §D for additional model details. For closed tools, as in Section 2, we test the OpenAI Moderation API and GPT-4.

[7]We experiment with various base models (see Appendix F.1 for comparisons). `Mistral-7b-v0.3` shows top performance, but difference margins are small, emphasizing primary importance of WILDGUARDTRAIN.

[8]We count XSTEST-RESP among public benchmarks because the prompts originate from XSTest [30].

Table 4: F1 scores (%) on WILDGUARDTEST. *Adv./ Vani.* indicate benign + harmful prompts that are adversarial and vanilla, respectively. *Harm.* indicates refusal detection on harmful prompts only.

Model	Prompt Harm.			Response Harm.			Refusal Detection			
	Adv.	Vani.	**Total**	Adv.	Vani.	**Total**	*Harm.*	Adv.	Vani.	**Total**
Llama-Guard	32.6	70.5	56.0	25.8	66.7	50.5	76.5	45.1	56.9	51.4
Llama-Guard2	46.1	85.6	70.9	47.9	78.2	66.5	82.2	47.9	58.8	53.8
Aegis-Guard-D	74.5	82.0	78.5	40.4	57.6	49.1	56.1	38.9	44.0	41.8
Aegis-Guard-P	62.9	77.9	71.5	49.0	62.4	56.4	67.1	45.4	48.2	46.9
HarmB-Llama	-	-	-	41.9	50.2	45.7	89.8	74.0	72.5	73.1
HarmB-Mistral	-	-	-	51.8	70.3	60.1	79.3	56.1	60.5	58.6
MD-Judge	-	-	-	67.7	**85.0**	76.8	85.7	50.6	59.4	55.5
BeaverDam	-	-	-	51.2	74.3	63.4	80.7	48.4	59.4	54.1
LibrAI-LongFormer-harm	-	-	-	61.7	64.2	63.2	79.2	61.7	62.7	62.3
LibrAI-LongFormer-ref	-	-	-	-	-	-	84.1	71.2	69.3	70.1
Keyword-based	-	-	-	-	-	-	67.8	67.0	65.9	66.3
OAI Mod. API	6.8	16.3	12.1	14.7	18.8	16.9	71.4	44.5	54.3	49.8
GPT-4	81.6	**93.4**	87.9	**73.6**	81.3	**77.3**	93.9	**91.4**	**93.5**	**92.4**
WILDGUARD	**85.5**	91.7	**88.9**	68.4	81.5	75.4	**94.0**	88.5	88.6	88.6

4.2 Results: WILDGUARD sets a new multi-task state-of-the-art

WILDGUARD performs best for prompt classification. Results are shown in Table 3 and 4. On both evaluation sets, none of the open baselines outperform GPT-4. However, WILDGUARD outperforms GPT-4 by 1.8% average F1 on public prompt harmfulness benchmarks, and on WGTEST by 1.1%. Particularly on the *adversarial* subset of prompts, WILDGUARD demonstrates a large improvement of 11.0% over the best open baseline and outperforms GPT-4 by 3.9%.

WILDGUARD matches baselines on response harmfulness. Results are also shown in Table 3 and 4. For response harmfulness, on public benchmarks we see that WILDGUARD outperforms all open baselines by at least 1.8%, and is again the only open model to outperform GPT-4 (on two out of four evaluations). On WGTEST, WILDGUARD performs within 3% of MD-Judge, the best open baseline, with WILDGUARD performing better on responses to adversarial prompts by 1%.

WILDGUARD substantially upgrades refusal detection. On full refusal evaluation, Table 4 shows that WILDGUARD outperforms LibrAI-LongFormer-ref—the only existing open model that explicitly classifies refusal—by 26.4%, and the strongest open baseline by 21.2%. WILDGUARD also closes the gap between open models and GPT-4, with a score within 4.1% of GPT-4.

As in §2, for the benefit of baseline models trained to detect response harmfulness rather than refusal, we also include refusal evaluation on a setting with harmful prompts only (*Harm.*). In that setting, we see that WILDGUARD outperforms all open baselines, beating the best open baseline by almost 5%, and outperforming MD-Judge—the strongest model on response harmfulness—by a margin of 9.7%. Additionally, in this setting WILDGUARD outperforms the GPT-4 judge.

Summary. Across all three tasks on previous benchmarks and WGTEST, WILDGUARD shows performance beating or on par with GPT-4, and substantially outperforming existing open models, with the single exception of MD-Judge on WGTEST response harmfulness, for which WILDGUARD is of the same caliber. This underscores the flexibility and practicality of WILDGUARD, as it can be applied simultaneously to all three tasks of prompt harmfulness, response harmfulness, and response refusal classification with superior accuracy. Notably, WILDGUARD excels in refusal classification with benign prompts, a capability that no previous open model supports with adequate performance.

4.3 WILDGUARD Ablation Results

Table 5 shows a series of ablations on WGTRAIN, as well as ablations on the multi-task setting comparing against single-task models for each task on the same training data (see Table 12 and 13 for formats). We compare scores on WGTEST for all tasks, the average F1 of public evaluations (see Table 3) for prompt and response harmfulness, and XSTEST-RESP for response refusal.

Table 5: Ablations of WGTRAIN showing the contributions of the dataset. Results from individual single-task models are combined under "Single-task Only".

Model	Prompt Harmfulness			Response Harmfulness			Refusal Detection		
	Public	WGTEST		Public	WGTEST		XSTEST	WGTEST	
	Avg. F1	Adv. F1	Total F1	Avg. F1	Adv. F1	Total F1	F1	Adv. F1	Total F1
WILDGUARDTRAIN	**86.1**	85.5	88.9	**82.4**	68.4	75.4	92.8	**88.5**	**88.6**
− `Synthetic Adv.`	84.6	77.1	84.4	78.6	67.8	73.9	94.1	88.3	87.6
− `Synthetic Vani.`	83.6	**85.7**	**89.8**	81.3	66.9	75.5	90.2	85.7	84.9
− IN-THE-WILD	83.9	84.6	88.7	81.9	**69.5**	**76.5**	93.0	88.0	**88.6**
− `Worker-written`	84.8	84.5	86.8	76.4	66.4	74.7	**94.9**	87.8	88.4
Single-task Only	86.0	84.2	89.2	76.4	67.2	74.8	93.3	87.0	87.3

Each major component of WGTRAIN helps to improve moderation performance. WGTRAIN is composed of four sources of harmful and benign prompts (See §3); Table 5 shows that each is essential for high performance across tasks. For instance, when removing adversarial data, scores on almost all test evaluations drop, with a decrease of over 8.4 F1 points on WGTEST adversarial prompts. On the other hand, if we exclude the synthetic vanilla component, both public prompt harm evaluations and WGTEST refusal detection drop by 2.5-3.7 F1 points. Without IN-THE-WILD data, performance drops on public evaluations across tasks, especially an F1 drop of 10.3 points on ToxicChat—which consists of prompts from in-the-wild human-LLM interactions. Finally, excluding existing annotator-written data is much worse on public prompt and response harmfulness tasks (loss of 1.3-6 F1) and performs worse on all tasks in WGTEST.

Multi-task training improves the model performance. We additionally compare WILDGUARD trained under the multi-task setting to models trained on individual tasks. On all tasks except refusal detection on XSTEST-RESP, the multi-task model outperforms single-task models, demonstrating that multi-task training makes WILDGUARD an efficient unified tool without individual task performance.

4.4 Demonstrating WILDGUARD as a Moderator in Human-LLM Interactions

As a practical demonstration, we test WILDGUARD and other tools in a simulated chat moderation use-case in tandem with a completion model, detecting harmful prompts and responses and inserting benign refusals in place of the model's original response when harm is detected. For test prompts we use the full WILDJAILBREAK (WJ) validation set [20], consisting of 2000 harmful and 250 benign adversarial prompts. We apply the moderation filters to Tulu-2-dpo-7B [17], comparing against a baseline of Tulu-2 safety-tuned on the WJ train set (Tulu-2 + `WJ`). We measure Attack Success Rate (ASR) for harmful prompts and Refusal To Answer (RTA) rate for benign prompts, using GPT-4 to label responses for compliance versus refusal for both tasks. When using WILDGUARD, Llama-Guard, and Aegis-Guard as moderators, we use prompt harmfulness and response harmfulness outputs for prompt and response components, respectively, while for MD-Judge we use response harmfulness output for the combined prompt and response input.

Table 6: ASR and RTA on WILDJAILBREAK eval set with classifier models filtering out harmful prompts and responses.

Filter \ Model	Tulu2+`WJ`	Tulu2-dpo
	(ASR (%, ↓) / RTA (%, ↓))	
✗	2.3/**1.2**	79.8/**0.0**
WILDGUARD	**0.7**/1.7	**2.4**/0.4
Llama-Guard2	1.9/1.6	53.1/0.8
Aegis-Guard-D	0.9/16.8	12.4/16.0
Aegis-Guard-P	1.7/4.0	32.7/3.6
MD-Judge	1.9/10.1	25.7/4.4

Table 6 shows that WILDGUARD yields the strongest performance both for refusing harmful jailbreak prompts and for avoiding over-refusal—for each model, it achieves lowest ASR while minimally increasing refusals to benign prompts. Specifically, we see that Tulu-2 combined with WILDGUARD filter shows significant improvement on ASR (79.8% to 2.4%) with minimal sacrifice on RTA (0.0% to 0.4%). We also see that using WILDGUARD as a filter performs similarly to the directly safety-tuned Tulu-2 + WJ. This shows that for use-cases where training is not feasible, WILDGUARD can serve as a highly effective safety filter embedded in an LLM interface at inference time. Additionally, even when using a safety-tuned model such as Tulu-2 + WJ, we can further reduce ASR with a comparatively small increase in RTA by using WILDGUARD as an additional filter.